

Few Shot Learning for BioDiversity Assessment Using UAV Imagery

Supervisors:

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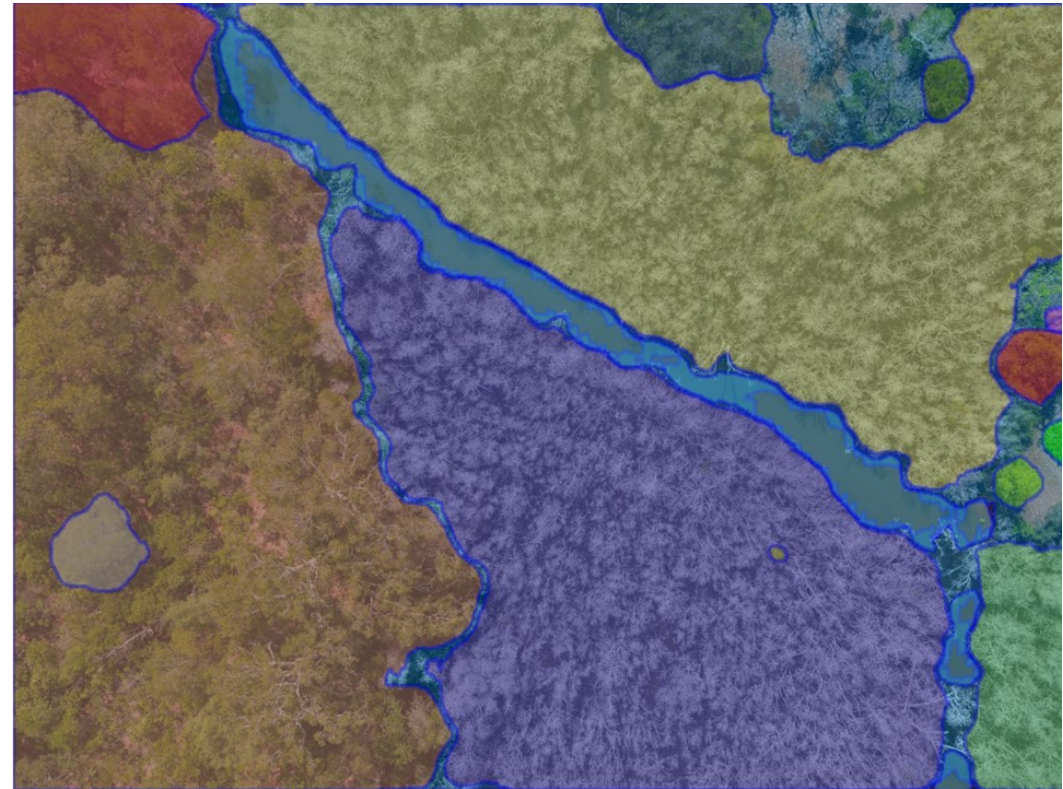
Ahmad Bdeir

BioDiversity Assessment

- Biodiversity loss is accelerating due to climate change and human activity
- Accurate monitoring is critical for conservation and policy decisions
- This is a global problem with different landscapes and greenery involved
 - Providing a few-shot model to help segment and classify land cover lessens the load on experts and policy makers

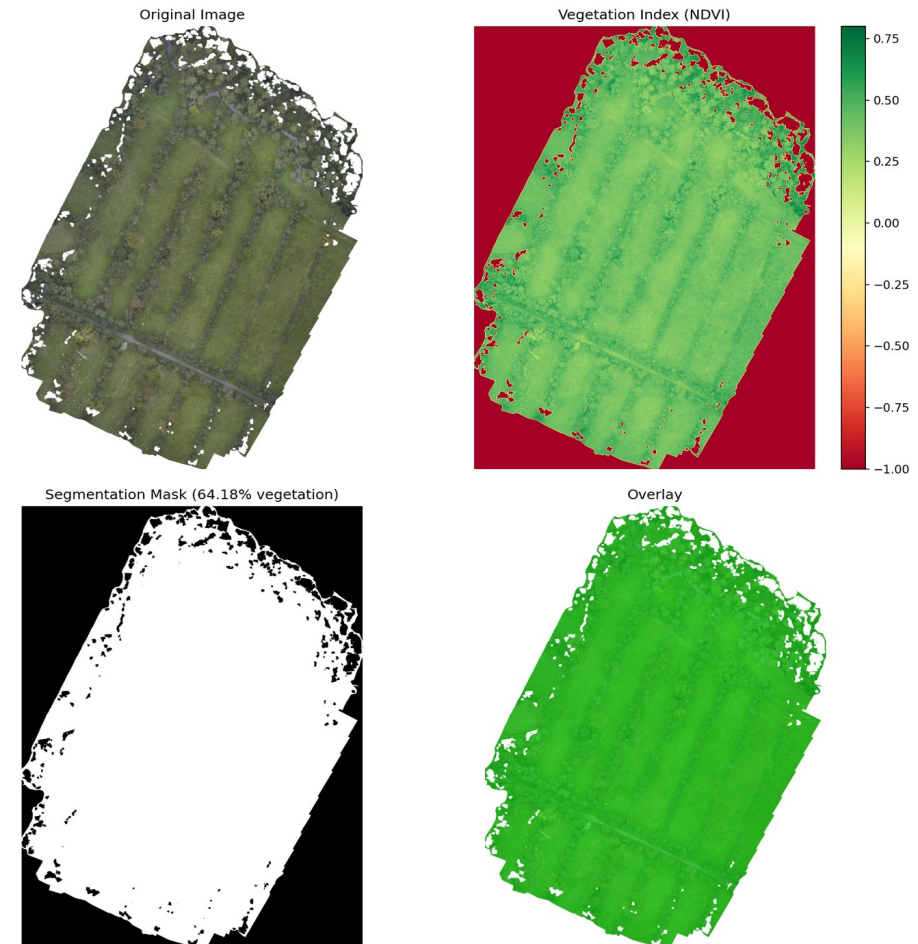
UAV Imagery in Biodiversity Monitoring

- High spatial resolution imagery
- Cost-effective and scalable
- Enables monitoring of:
 - Vegetation types
 - Species distribution
 - Habitat structure
- Sensors can be:
 - RGB
 - Multispectral
 - Thermal
 - HyperSpectral



Dataset

- EU Collected Data in multiple countries including:
 - Portugal
 - Germany
 - UK
 - Austria
- Sensor types:
 - RGB
 - Some have NIR as well



Approach

- The project would then be divided into phases:
 1. Research of available methodologies for segmentation models with a focus on the Segment Anything Model (SAM)
 2. Set up of the baseline models
 3. Adaptation of baseline models to Few-shot learning
 4. Integration of additional channels and heuristic masks (based on NDVI and Shannon Diversity Indices)
- Phase completion depends on the progress of the team